

NIFTY100 Financial Intelligence

Analyst User Guide

Purpose: This guide explains how analysts can use the financial dashboard, company profiles, screener, peer analysis, valuation information, API services and generated tearsheet reports.

Project: NIFTY100 Financial Intelligence

Audience: Financial analysts, research users and project reviewers

Version: Sprint 6 Final

1. Project Overview

The NIFTY100 Financial Intelligence application provides a centralized interface for analysing company financial information. The system combines cleaned financial datasets, calculated KPIs, screening filters, peer comparisons, valuation history, clustering and downloadable reports.

Main capabilities

- Browse company information and latest financial KPIs.
- Filter companies using financial screening parameters.
- View historical Profit & Loss, Balance Sheet and Cash Flow information.
- Analyse calculated financial ratios and growth metrics.
- Compare companies with their peer groups.
- Review historical valuation multiples.
- Generate and download company tearsheet PDFs.
- Access the same analytical data through FastAPI endpoints.

Architecture

The project uses a data-processing layer, SQLite database, analytical calculations, FastAPI backend services and a Streamlit dashboard. The API provides structured JSON responses while the dashboard provides an analyst-friendly interface.

2. Running the Application

Start FastAPI

Open a terminal in the project root and run:

```
uvicorn src.api.main:app --port 8000
```

The FastAPI service should become available at port 8000.

Open API documentation

Navigate to the FastAPI documentation page:

```
http://127.0.0.1:8000/docs
```

Start Streamlit

Open a second terminal and start the dashboard using the project's configured Streamlit entry point. A typical command is:

```
streamlit run app.py
```

The dashboard normally runs on port 8501.

Important

- Start FastAPI before using dashboard features that depend on the API.
- Keep FastAPI and Streamlit on different ports.
- Do not close the terminal running either service.

3. Using the Streamlit Screener

The screener allows analysts to narrow the company universe using financial criteria. The results are ranked and display the relevant financial metrics.

Common screening parameters

Parameter	Purpose
min_roe	Minimum Return on Equity
max_de	Maximum Debt-to-Equity
min_fcf	Minimum Free Cash Flow
sector	Filter by sector
min_rev_cagr_5yr	Minimum 5-year revenue CAGR
min_pat_cagr_5yr	Minimum 5-year PAT CAGR
max_pe	Maximum Price/Earnings ratio

Enter one or more filters and apply the screener. Review the returned company list and verify that the displayed metrics satisfy the selected conditions.

4. Company Profile

The Company Profile screen provides a detailed view of an individual company.

Profile information

- Company identification and sector information.
- Latest year financial KPIs.
- Historical financial information.
- Profit & Loss history.
- Balance Sheet history.
- Cash Flow history.
- Calculated ratios and growth metrics.
- Peer and valuation information.

Opening a company

Select a company from the screener or navigate directly using its ticker. The profile endpoint also supports direct retrieval using the company ticker.

Example API request:

```
GET /api/v1/companies/TCS
```

If a ticker does not exist, the API returns HTTP 404.

Analyst usage

Use the company profile to inspect individual financial trends before making comparisons or applying a screening decision.

5. Financial Analysis Screens

Profit & Loss

The P&L view presents historical revenue, profitability and related financial values. Year filters can be used to restrict the displayed period.

Balance Sheet

The Balance Sheet view provides historical assets, liabilities, equity and related financial information.

Cash Flow

The Cash Flow view provides historical cash-flow information and helps analysts assess operating, investing and financing cash movements.

Ratios

The ratios endpoint provides calculated KPIs for each available year. Important measures include ROE, Debt-to-Equity, Interest Coverage Ratio, CAGR measures and Operating Profit Margin.

Example:

```
GET /api/v1/companies/TCS/ratios
```

A year parameter can be supplied when a single year's KPI values are required.

6. Peer Comparison and Valuation

Peer groups

The peer analysis screen allows an analyst to compare companies belonging to the same peer group. Percentile ranks are available for the core financial metrics.

Example:

```
GET /api/v1/peers/IT
```

Radar comparison

Company comparison provides multiple financial axes for the selected company, peer-group average and benchmark company. This helps identify relative strengths and weaknesses.

Historical valuation

The market-cap valuation endpoint provides historical valuation multiples including P/E, P/B, EV/EBITDA and dividend yield.

Example:

```
GET /api/v1/market-cap/TCS
```

7. Tearsheets and Downloads

The application provides pre-generated company tearsheet PDFs containing key financial and analytical information.

Downloading a tearsheet

From a company profile, use the available tearsheet/download action. The API also exposes the tearsheet as a binary PDF response.

Example:

```
GET /api/v1/companies/TCS/tearsheet
```

Expected output

- Content type: application/pdf
- One tearsheet PDF per company.
- PDF should contain readable financial information.
- The final acceptance check verifies that sampled PDFs do not contain text overflow.

Analyst recommendation

Use the tearsheet when a compact company-level report is required for offline review or sharing.

8. FastAPI Usage

The backend exposes versioned REST endpoints under the `/api/v1` prefix.

Health check

`GET /api/v1/health`

The health endpoint reports API status, database row counts, uptime and application version.

Companies

`GET /api/v1/companies`

Returns the company universe with optional filters such as sector, market-cap category and search.

Screener

`GET /api/v1/screener?min_roe=15`

Sectors

`GET /api/v1/sectors`

`GET /api/v1/sectors/IT/companies`

Documents

`GET /api/v1/companies/TCS/documents`

The documents endpoint returns annual-report links together with URL validity information.

9. API Examples with cURL

The following examples can be executed from a terminal when the FastAPI server is running.

Health

```
curl http://127.0.0.1:8000/api/v1/health
```

Company list

```
curl http://127.0.0.1:8000/api/v1/companies
```

Company profile

```
curl http://127.0.0.1:8000/api/v1/companies/TCS
```

Ratios

```
curl http://127.0.0.1:8000/api/v1/companies/TCS/ratios
```

Screener

```
curl "http://127.0.0.1:8000/api/v1/screener?min_roe=15&max_pe=30"
```

API documentation

The interactive OpenAPI documentation is available at /docs and can be used to inspect request parameters, response schemas and available operations.

10. Troubleshooting

FastAPI does not start

- Check whether port 8000 is already being used.
- Confirm that the project virtual environment is active.
- Check the terminal error for missing dependencies or import errors.

Dashboard cannot load API data

- Confirm FastAPI is running.
- Verify the API base URL and port.
- Open the health endpoint to confirm database connectivity.

Company returns 404

- Check the ticker spelling.
- Confirm the ticker exists in the company table.

Slow response

- Check the performance notes generated during Sprint 6.
- Review SQLite queries and indexes.
- Avoid unnecessary repeated database queries.

PDF generation problem

- Verify the source company data exists.
- Check the reports/tearsheets directory.
- Confirm generated PDF files are non-empty and readable.

11. Data Quality and Analyst Checks

Analysts should verify that dashboard results are consistent with the underlying API and database data.

- Company count should be 92.
- The project contains 11 sectors.
- Financial history should cover the expected years for each company.
- Calculated KPIs should follow the project's defined formulas.
- Screening results should satisfy every selected filter.
- Peer results should belong to the requested peer group.
- Downloaded CSV files should be valid and well formed.
- Generated tearsheet PDFs should open correctly.

Performance checks

Sprint 6 performance testing measures concurrent screener API requests and Company Profile response times. The detailed measurements are stored in output/perf_notes.md.

Final analyst workflow

A recommended workflow is: start the services → open the dashboard → use the screener → select a company → review financial history → inspect ratios → compare peers → review valuation → download the tearsheet.

12. Final Reference

This guide provides the operational instructions required for analysts to use the NIFTY100 Financial Intelligence application.

Important locations

- src/api/ — FastAPI application and routers
- output/ — generated analytical and validation outputs
- reports/ — charts, reports and generated analysis
- reports/tearsheets/ — company tearsheet PDFs
- docs/ — project documentation
- tests/ — automated tests

Key endpoints

- GET /api/v1/health
- GET /api/v1/companies
- GET /api/v1/companies/{ticker}
- GET /api/v1/companies/{ticker}/pl
- GET /api/v1/companies/{ticker}/bs
- GET /api/v1/companies/{ticker}/cashflow
- GET /api/v1/companies/{ticker}/ratios
- GET /api/v1/screener
- GET /api/v1/sectors
- GET /api/v1/peers/{group_name}
- GET /api/v1/market-cap/{ticker}

End of Analyst Guide